

3. Qualitative analysis is used to identify the ions present in a compound.

- (a) The table shows the observations from chemical tests carried out on solutions of two compounds.

Sample	Test	Observation	Name of compound tested
1	add dilute acid	colourless gas evolved which turns limewater milky	
	flame test	lilac	
2	flame test	lilac	
	add nitric acid then silver nitrate	white precipitate	

Complete the table.

[3]

- (b) Another compound was identified as calcium sulfate (CaSO_4).
Use the information below and the equation:

$$\text{number of moles} = \frac{\text{mass}}{M_r}$$

to calculate the number of moles of calcium sulfate present in 34 g of powder.

[4]

Element	A_r
Calcium	40
Sulfur	32
Oxygen	16

number of moles =

